GRADE 7 ACCELERATED · UNIT 4 · ASSESSMENT

Solving Problems with Rational Numbers

34 points

Name_____
Date_____
Period_____

Show your work. Circle your final answer.

This assessment runs over two class periods. Stop when the period ends; you will continue from where you stopped.

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. Operations in Scientific Notation

12 points

1. Add or subtract. Write each answer in scientific notation.

a. $(3.6 \times 10^5) + (2.1 \times 10^5)$ *1 pt*

b. $(8.4 \times 10^{-6}) - (5.9 \times 10^{-6})$ *1 pt*

c. $(7.5 \times 10^8) + (6.8 \times 10^8)$ *1 pt*

d. $(4.2 \times 10^4) + (3.7 \times 10^3)$ *1 pt*

2. Multiply or divide. Write each answer in scientific notation.

a. $(3 \times 10^4)(2 \times 10^6)$ *1 pt*

b. $\frac{9.6 \times 10^9}{1.2 \times 10^4}$ *1 pt*

c. $(5 \times 10^{-3})(4 \times 10^7)$ *1 pt*

d. $\frac{1.8 \times 10^2}{6 \times 10^5}$ *1 pt*

3. Which is the value of $(1.4 \times 10^3) + (2.9 \times 10^5)$? *1 pt*

- ○ A. 3.04×10^5
- ○ B. 4.3×10^5
- ○ C. 2.914×10^5
- ○ D. 4.3×10^8

4. Select ALL of the expressions whose value is 3×10^6 . *1 pt*

- ☐ A. $(3.2 \times 10^6) - (2 \times 10^5)$
- ☐ B. $\frac{9 \times 10^9}{3 \times 10^4}$
- ☐ C. $(6 \times 10^3)(5 \times 10^2)$
- ☐ D. $(1.8 \times 10^6) + (1.2 \times 10^6)$
- ☐ E. $(1.8 \times 10^6) + (1.2 \times 10^5)$
- ☐ F. $\frac{9 \times 10^9}{3 \times 10^3}$

5. Kendra writes $(2.3 \times 10^4) + (6.1 \times 10^6) = 8.4 \times 10^{10}$. Name the two mistakes she made, and give the correct sum. *1 pt*

6. A number is multiplied by 4×10^3 and the product is 2×10^{-1} . What was the number? Write your answer in scientific notation. *1 pt*

2. Real-World Problems and Significant Digits *9 points*

7. Give the number of significant digits in each measurement.

a. 3,080 grams *1 pt*

b. 0.00405 meters *1 pt*

c. 12.500 seconds *1 pt*

8. Solve. Report each answer with the correct number of significant digits.

a. A rectangular plate measures 2.4×10^2 mm by 1.35×10^2 mm. Find its area. *1 pt*

b. A vehicle travels 5.60×10^2 km in 8.0 hours. Find its average speed. *1 pt*

9. Solve. Report each answer with the correct precision.

a. Two lengths are measured: 6.25×10^{3} mm and 4.1×10^{2} mm. Find their sum. *1 pt*

b. Two masses are measured: 9.30×10^{-2} g and 4.72×10^{-3} g. Find their difference. *1 pt*

10. Jonah adds two measured lengths, 4,560 m and 3.2 m, and says the sum must be written with two significant digits, as 4.6×10^{3} m. Explain why he is wrong, and give the correct sum. *1 pt*

11. A rectangle's sides are measured as 8.0×10^{1} cm and 3.15×10^{1} cm. Which is the area, in scientific notation with the correct number of significant digits? *1 pt*

- ○ A. 2.5×10^{3} cm^2
- ○ B. 2.52×10^{3} cm^2
- ○ C. 2.5×10^{2} cm^2
- ○ D. 25.2×10^{2} cm^2

3. The Order of Operations with Exponents and Radicals *13 points*

12. Evaluate each expression.

a. $15 - 2 \cdot 3^2$ *1 pt*

b. $\frac{5^2 + 7}{2^3}$ *1 pt*

c. $36 \div 4 \cdot 3$ *1 pt*

d. $-4^2 + (-4)^2$ *1 pt*

13. Evaluate each expression.

a. $\sqrt{6^2 + 13}$ *1 pt*

b. $\sqrt[3]{-125} + 2^{-2}$ *1 pt*

c. $\left(-\frac{1}{3}\right)^2 + \sqrt{\frac{4}{9}}$ *1 pt*

14. A square patio of side 14 feet is sealed at 2.40 dollars a square foot, with a flat 85 dollars for delivery.

a. Write an expression for the total cost. *1 pt*

b. Find the total cost. *1 pt*

15. Which is the value of $2 + 5 \cdot 3^2$? *1 pt*

- ○ A. 32
- ○ B. 47
- ○ C. 63
- ○ D. 289

16. Select ALL of the expressions whose value is 6. *1 pt*

- ☐ A. $-\sqrt{36}$
- ☐ B. $\sqrt{36}$
- ☐ C. $36 \div 3 \cdot 2$
- ☐ D. $\left(\frac{1}{6}\right)^{-1}$
- ☐ E. $\sqrt{5^2 + 11}$
- ☐ F. $\frac{4^2 + 8}{4}$

17. Nadia writes $\sqrt{25+144} = \sqrt{25} + \sqrt{144} = 17$. Explain why she is wrong, and give the correct value. *1 pt*

18. A square tile has an area of 169 square inches. A second square tile has a side twice as long as the first. What is the area of the second tile? *1 pt*